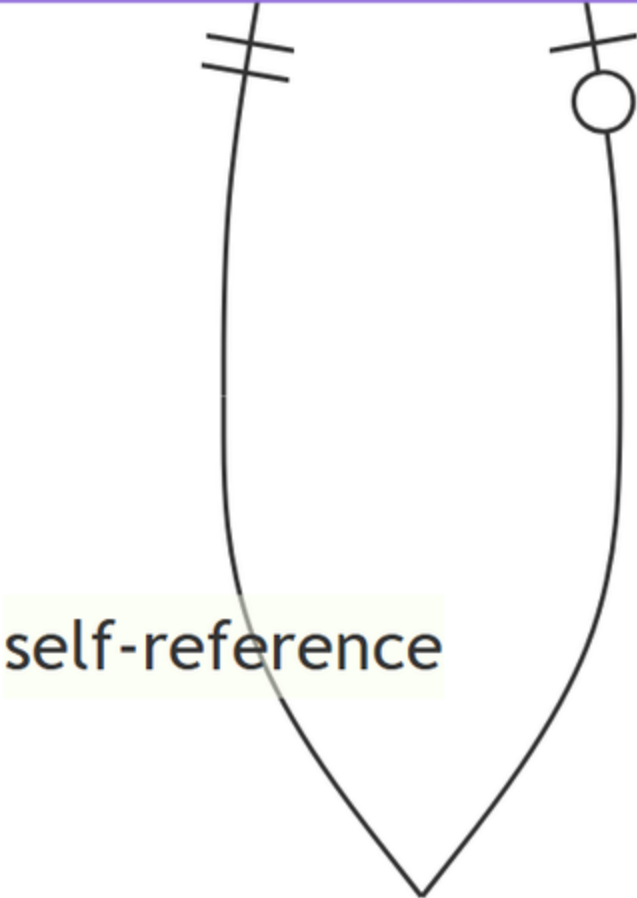


1. URL Shortener Data Model

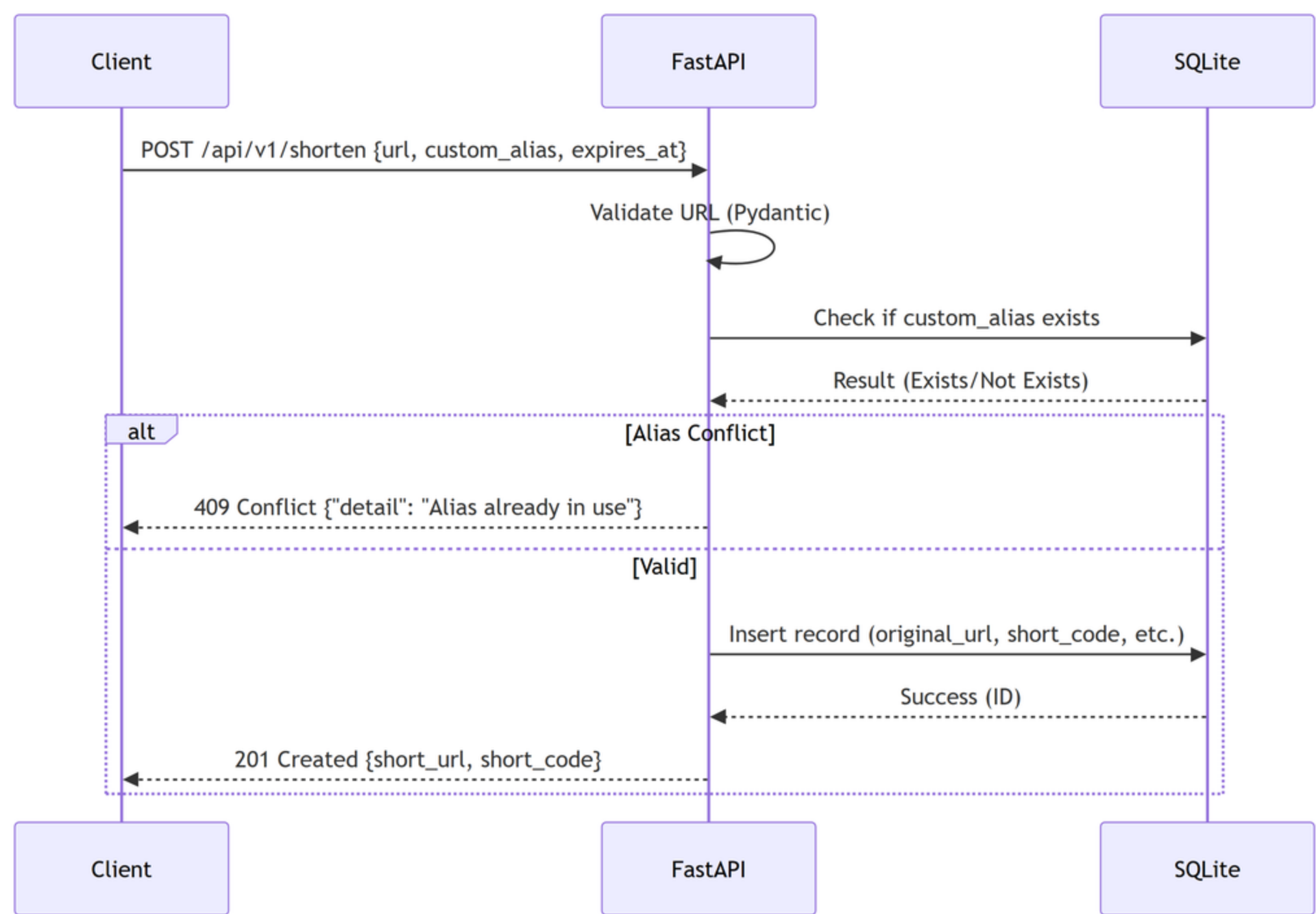
Entity Relationship diagram for the URLRecord entity in SQLite, detailing attributes and constraints.

URLRECORD			
int	id	PK	
string	original_url		
string	short_code		Unique
string	custom_alias		Unique, Optional
int	clicks_count		
datetime	created_at		
datetime	expires_at		Optional
boolean	is_active		



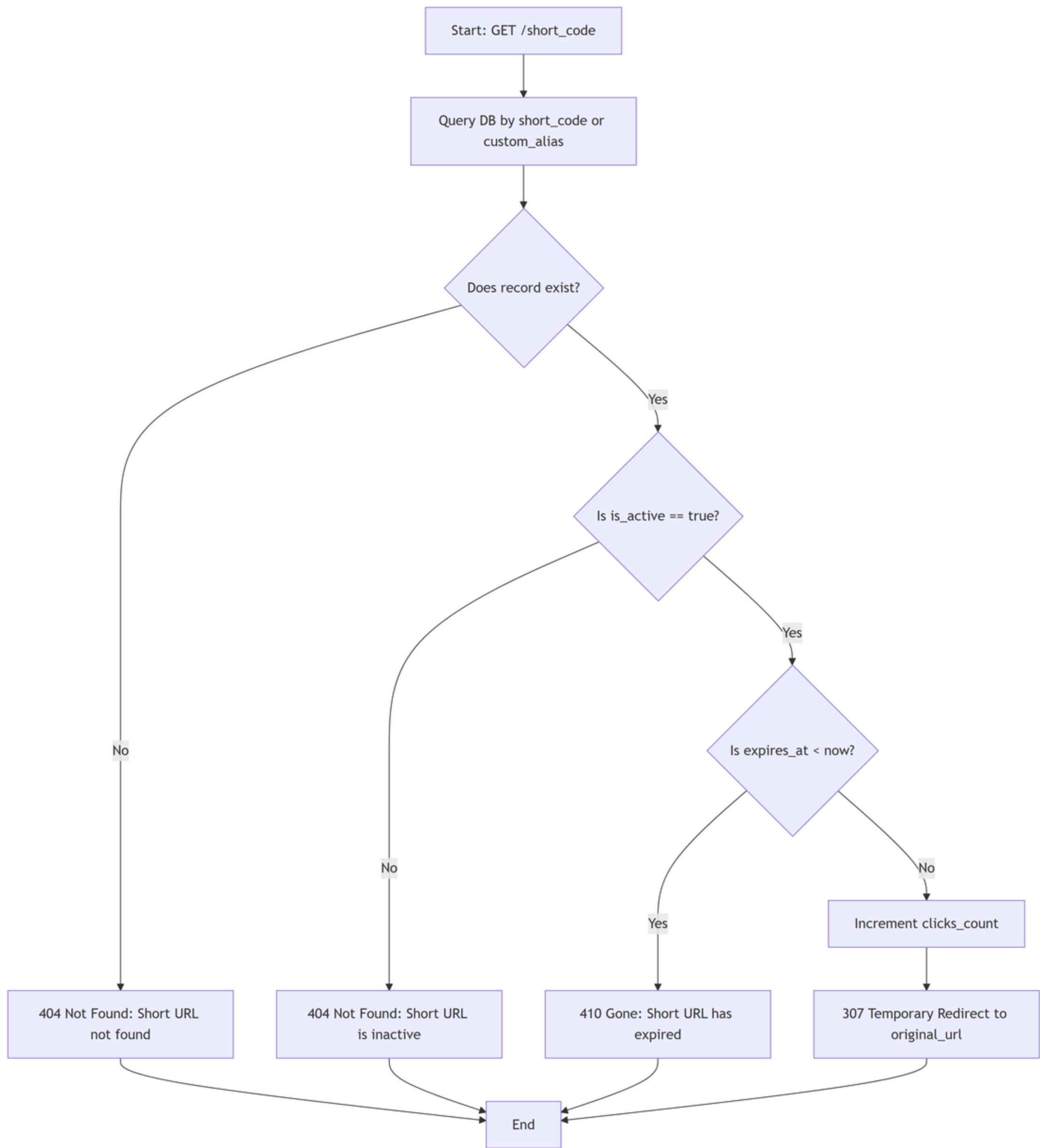
2. Short URL Creation Sequence

Detailed interaction flow for creating a shortened URL, including validation and conflict handling.



3. URL Redirection Logic Flow

Decision logic for handling redirection requests, including expiration and activity checks.



4. Requirements Traceability Matrix

Mapping of User Stories to Functional Requirements and the underlying Data Entity.

